

PROFESSIONAL SUMMARY

- Python developer with 5+ years building and maintaining production backend APIs, data pipelines, and ML-integrated services — using FastAPI, asyncio, and Python scripting across cloud and Linux environments.
- Full-stack capable: React, Angular, and Node.js frontend paired with Python/FastAPI and REST API backends, with hands-on experience in RESTful design, OAuth 2.0 / JWT auth, and IAM security patterns.
- Applied ML and AI experience: pandas, NumPy, scikit-learn, MLflow model lifecycle, pgvector embeddings, IPython/Jupyter for interactive experimentation, and prompt design and evaluation with Anthropic/OpenAI APIs.
- Strong collaborator and code quality advocate — experienced in Agile teams, Git branching workflows, peer code reviews, and writing clean, well-documented, maintainable code.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Go, SQL, Bash

Python Frameworks & Libraries: FastAPI, asyncio, pandas, NumPy, scikit-learn, MLflow, IPython / Jupyter

Frontend: React, Angular, Node.js, Express.js, Material UI, HTML5, CSS3, SSR

IAM & Security: OAuth 2.0, JWT, RBAC, Azure AD / Active Directory, SSO, REST API authentication and authorisation

AI & GenAI: Anthropic / OpenAI APIs, Model Context Protocol (MCP), pgvector (RAG pipelines), prompt design and AI output evaluation, GitHub Copilot

Databases: PostgreSQL, MySQL, MongoDB, SQL Server — data modeling, query optimisation, ORM

DevOps & Cloud: AWS (EC2, S3), Azure (VM, Service Bus, Blob Storage), Docker, Kubernetes, GitHub Actions, CI/CD, Linux / Unix

EXPERIENCE

Senior Software Engineer | 04/2023 – 12/2023 | Tecnotree | Bengaluru, KA

Awarded "Spotlight" cash award as top-performing engineer for shipping production-quality Python APIs and React features with minimal post-release defects.

- Built and maintained RESTful API services using Python and FastAPI, delivering backend features consumed by the React frontend — writing clean, well-documented code following team coding standards.
- Enforced software quality through Jest and Python unit/integration testing achieving 90%+ coverage across API contracts and UI components, reducing regression risk across iterative Agile releases.
- Participated in Git pull request workflows, peer code reviews, and JIRA story management — collaborating closely with product and design to translate requirements into technical specifications.

Senior Software Engineer | 07/2022 – 03/2023 | PangeaTech | Bengaluru, KA

Founding engineer on a production ML SaaS platform — owned the Python backend, IAM security layer, AI/ML pipeline, and cloud infrastructure design from inception through production.

- Designed and deployed Python FastAPI microservices with OAuth 2.0 / JWT authentication and Azure AD integration, implementing RBAC and SSO-compatible authorisation patterns across all API endpoints.
- **ML pipeline:** Built and managed end-to-end ML model lifecycle using MLflow — experiment tracking, model versioning, and deployment governance — with IPython / Jupyter notebooks for interactive data exploration and validation.
- **RAG & vector embeddings:** Implemented retrieval-augmented generation pipelines using PostgreSQL pgvector and Python, enabling semantic search and AI-driven context retrieval across enterprise datasets.
- **AI output evaluation:** Designed prompt templates and structured evaluation frameworks using Anthropic APIs, assessing output quality, factual grounding, and edge-case robustness across AI-integrated features.
- **Real-time observability:** Deployed centralised logging via the EFK stack on Linux servers — writing Bash and Python scripts to automate log rotation, process monitoring, and routine operational tasks.
- **30% cloud cost reduction:** Optimised Node.js and Python process scaling on Azure, cutting cloud spend 30% while maintaining SLA uptime commitments.
- Authored 50+ code reviews and mentored junior engineers on Python best practices, security patterns, and clean code principles in a globally distributed Agile/Scrum team.

Software Engineer | 11/2021 – 07/2022 | PangeaTech | Bengaluru, KA

- **97% DB performance gain:** Engineered a 97% reduction in database refresh time (8 hrs → 15 min) by refactoring synchronous Python to fully asynchronous Python (asyncio) — the flagship example of hands-on Python optimisation.
- **85% faster deployments:** Containerised Python microservices with Docker, orchestrated with Kubernetes, and automated CI/CD via GitHub Actions — eliminating deployment failures and cutting release time from 27 min to 4 min.
- Designed and maintained 50+ PostgreSQL materialized views powering real-time client dashboards — applying ORM patterns, query optimisation, and data integrity controls at scale.
- Managed and troubleshooted application services on Linux/Unix environments — monitoring logs, managing processes, and writing Python and Bash automation scripts to streamline operations.

Software Engineer | 07/2019 – 04/2021 | Tenable (a JPMorgan Chase company) | Pune, MH

- Developed Python and TypeScript services for an enterprise travel-booking platform serving millions of users — applying TDD, SOLID design principles, and RESTful API best practices throughout.
- Achieved 90% unit test coverage across components, ensuring WCAG accessibility compliance and secure, authenticated user flows across integrated third-party services.

PROJECTS & ADDITIONAL EXPERIENCE

Travel AI Agent (MCP) | Personal Project | Python-based agentic AI tool

- Built a Python agentic reasoning system using Model Context Protocol (MCP) and Anthropic APIs — designing prompt templates, evaluating AI output quality, and iterating on grounding and reliability across multi-step workflows.

Email Marketing Microservice | ABC Academy of Music | Python API platform

- Led a team of 5 engineers to design and ship a Python-backed email campaign platform — driving API architecture decisions, Agile sprint planning, and collaboration with the product owner.
- Boosted team productivity by 75% by introducing AI-assisted development workflows (GitHub Copilot) as part of an engineering-wide adoption initiative.

Python Board Game | University Capstone

- Awarded university-level "Best Paper" for the technical design of a turn-based strategy game built entirely in Python — demonstrating strong algorithmic problem-solving, object-oriented design, and clean code principles.

Engineering Tutor | ASU: Delivered 100+ tutoring sessions on Data Structures and Algorithms, Operating Systems, and Database Management.

EDUCATION

Master of Science in Software Engineering, GPA: 3.87 / 4.0 | Dec 2025 | Arizona State University | Mesa, AZ

Coursework: System Design, Data Structures and Algorithms, User Experience, Agile Software Development

Bachelor of Science in Computer Science | Aug 2019 | REVA University | Bengaluru, KA